

Features

- $BV_{CEO} > 25V$
- $I_C = 50mA$ High Continuous Collector Current
- Saturation Voltage $< 500mV$ $I_C = 4mA$
- 350mW Power Dissipation
- **Totally Lead-Free & Fully RoHS Compliant (Notes 1 & 2)**
- **Halogen and Antimony Free. "Green" Device (Note 3)**
- **An automotive-compliant part is available under a separate datasheet ([MMBTH10Q](#))**

Mechanical Data

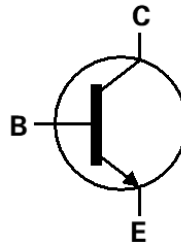
- Package: SOT23
- Package Material: Molded Plastic, "Green" Molding Compound.
UL Flammability Classification Rating 94V-0
- Moisture Sensitivity: Level 1 per J-STD-020
- Terminals: Finish – Matte Tin Plated Leads, Solderable per
MIL-STD-202, Method 208 (B3)
- Weight 0.008 grams (Approximate)

Applications

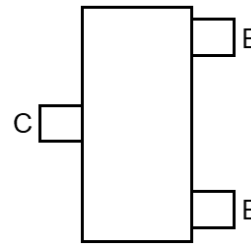
- VHF/UHF amplifiers
- High output VHF oscillators
- RF amplifiers



Top View



Device Symbol



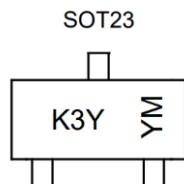
Top View
Pin Configuration

Ordering Information (Note 4)

Orderable Part Number	Package	Marking	Reel Size (inches)	Tape Width (mm)	Packing	
					Qty.	Carrier
MMBTH10-13	SOT23	K3Y	7	8	3,000	Reel

- Notes:
1. No purposely added lead. Fully EU Directive 2002/95/EC (RoHS), 2011/65/EU (RoHS 2) & 2015/863/EU (RoHS 3) compliant.
 2. See <https://www.diodes.com/quality/lead-free/> for more information about Diodes Incorporated's definitions of Halogen- and Antimony-free, "Green" and Lead-free.
 3. Halogen- and Antimony-free "Green" products are defined as those which contain <900ppm bromine, <900ppm chlorine (<1500ppm total Br + Cl) and <1000ppm antimony compounds.
 4. For packaging details, go to our website at <https://www.diodes.com/design/support/packaging/diodes-packaging/>.

Marking Information



K3Y = Product Type Marking Code
YM = Date Code Marking
Y = Year (ex: N = 2026)
M = Month (ex: 9 = September)

Date Code Key

Year	2001	-	2026	2027	2028	2029	2030	2031	2032	2033	2034	2035
Code	M	-	N	P	R	S	T	U	V	W	X	Y

Month	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Code	1	2	3	4	5	6	7	8	9	O	N	D

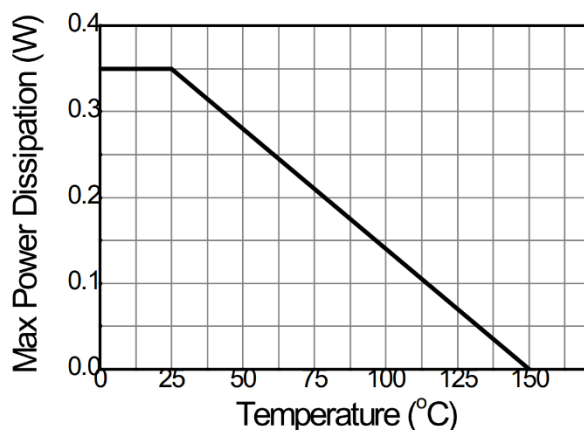
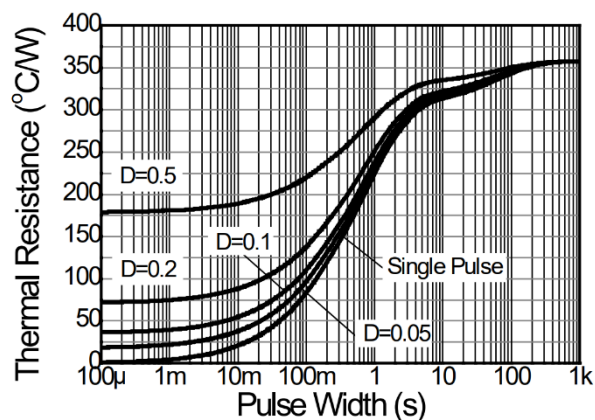
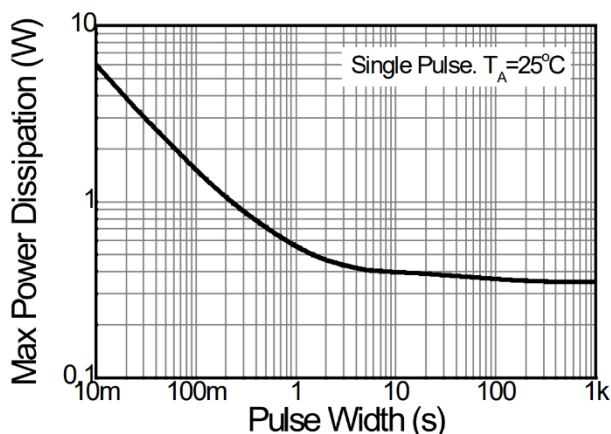
Absolute Maximum Ratings ($T_A = +25^\circ\text{C}$, unless otherwise specified.)

Characteristic	Symbol	Value	Unit
Collector-Base Voltage	V_{CB0}	30	V
Collector-Emitter Voltage	V_{CEO}	25	V
Emitter-Base Voltage	V_{EBO}	3	V
Collector Current	I_C	50	mA

Thermal Characteristics ($T_A = +25^\circ\text{C}$, unless otherwise specified.)

Characteristic	Symbol	Value	Unit
Power Dissipation	P_D	310	mW
		350	
Thermal Resistance, Junction to Ambient	$R_{\theta JA}$	403	$^\circ\text{C/W}$
		357	
Thermal Resistance, Junction to Leads	$R_{\theta JL}$	350	$^\circ\text{C/W}$
Operating and Storage Temperature Range	T_J, T_{STG}	-65 to +150	$^\circ\text{C}$

Notes: 5. For a device mounted on minimum recommended pad layout FR-4 PCB with high coverage of single-sided 1oz copper; device is measured under still air conditions whilst operating in a steady state.
 6. Same as Note 5, except mounted on 15mm x 15mm 1oz copper.
 7. Thermal resistance from junction to solder-point (at the end of the collector lead).

Thermal Characteristics and Derating Information

Fig.1 Derating Curve

Fig.2 Transient Thermal Impedance

Fig.3 Pulse Power Dissipation

Electrical Characteristics ($T_A = +25^\circ\text{C}$, unless otherwise specified.)

Characteristic	Symbol	Min	Max	Unit	Test Condition
OFF CHARACTERISTICS (Note 8)					
Collector-Emitter Breakdown Voltage	BV_{CEO}	25	—	V	$I_C = 1\text{mA}$, $I_B = 0$
Collector-Base Breakdown Voltage	BV_{CBO}	30	—	V	$I_C = 100\mu\text{A}$, $I_E = 0$
Emitter-Base Breakdown Voltage	BV_{EBO}	3	—	V	$I_E = 10\mu\text{A}$, $I_C = 0$
Collector Cutoff Current	I_{CBO}	—	100	nA	$V_{CB} = 25\text{V}$, $I_E = 0$
Emitter Cutoff Current	I_{EBO}	—	100	nA	$V_{EB} = 2\text{V}$, $I_C = 0$
ON CHARACTERISTICS (Note 8)					
DC Current Gain	h_{FE}	60	—	—	$I_C = 4\text{mA}$, $V_{CE} = 10.0\text{V}$
Collector-Emitter Saturation Voltage	$V_{CE(sat)}$	—	0.5	V	$I_C = 4\text{mA}$, $I_B = 400\mu\text{A}$
Base-Emitter On Voltage	$V_{BE(on)}$	—	0.95	V	$I_C = 4\text{mA}$, $V_{CE} = 10.0\text{V}$
SMALL-SIGNAL CHARACTERISTICS					
Current Gain-Bandwidth Product	f_T	650	—	MHz	$V_{CE} = 10\text{V}$, $f = 100\text{MHz}$, $I_C = 4\text{mA}$
Collector-Base Capacitance	C_{IBO}	—	0.7	pF	$V_{CB} = 10\text{V}$, $f = 1.0\text{MHz}$, $I_E = 0$
Collector-Base Feedback Capacitance	C_{OBO}	—	0.65	pF	$V_{CB} = 10\text{V}$, $f = 1.0\text{MHz}$, $I_E = 0$
Collector-Base Time Constant	$R_b'C_c$	—	9	ps	$V_{CB} = 10\text{V}$, $f = 31.8\text{MHz}$, $I_C = 4\text{mA}$

Note: 8. Measured under pulsed conditions. Pulse width $\leq 300\mu\text{s}$. Duty cycle $\leq 2\%$.

Typical Electrical Characteristics ($T_A = +25^\circ\text{C}$, unless otherwise specified.)

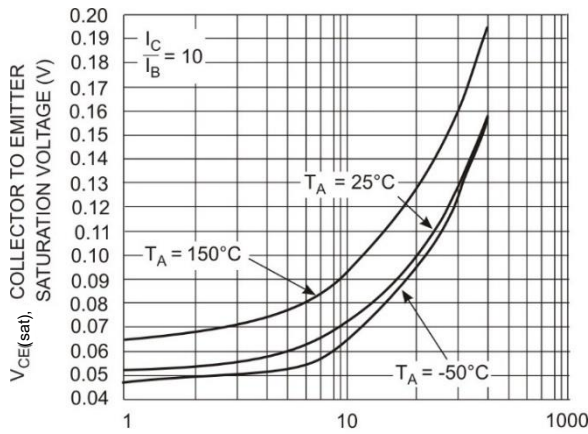


Fig.4 Collector Emitter Saturation Voltage vs. Collector Current

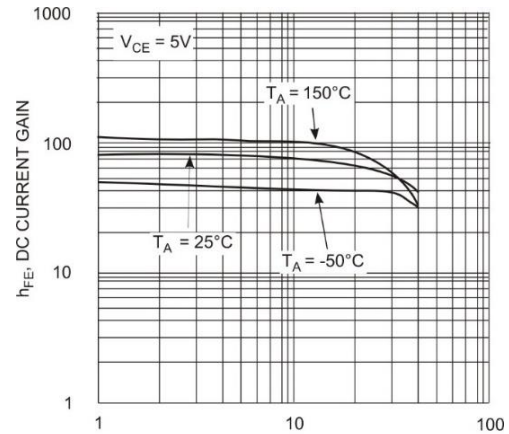


Fig.5 DC Current Gain vs. Collector Current

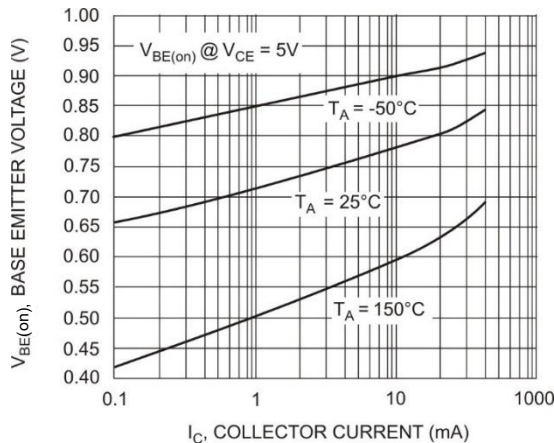


Fig.6 Base Emitter Voltage vs. Collector Current

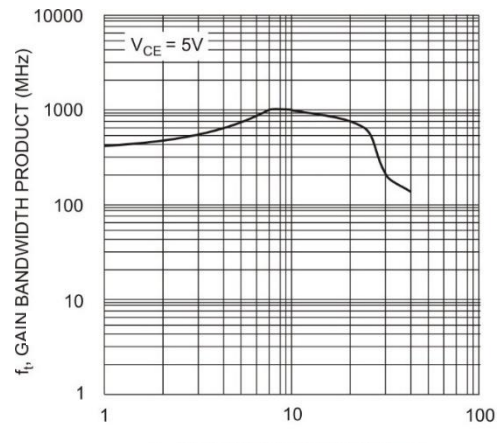
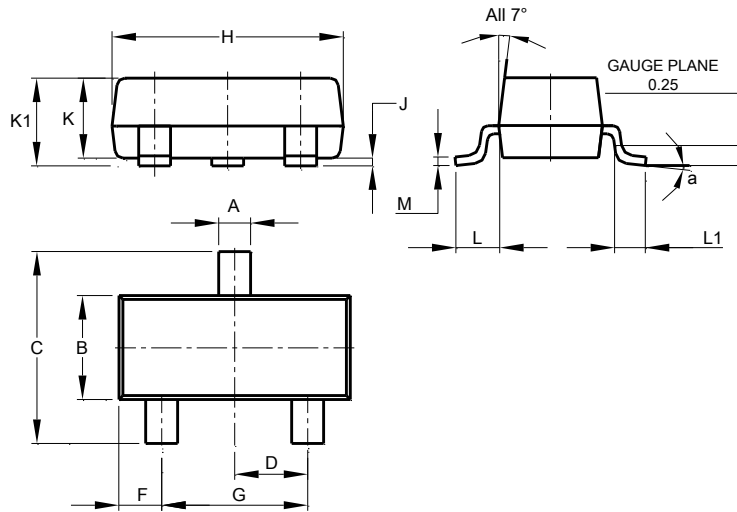


Fig. 7 Gain Bandwidth Product vs Collector Current

Package Outline Dimensions

Please see <http://www.diodes.com/package-outlines.html> for the latest version.

SOT23

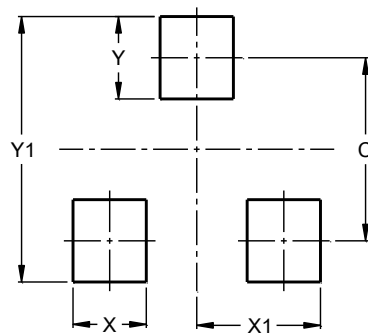


SOT23			
Dim	Min	Max	Typ
A	0.37	0.51	0.40
B	1.20	1.40	1.30
C	2.30	2.50	2.40
D	0.89	1.03	0.915
F	0.45	0.60	0.535
G	1.78	2.05	1.83
H	2.80	3.00	2.90
J	0.013	0.10	0.05
K	0.890	1.00	0.975
K1	0.903	1.10	1.025
L	0.45	0.61	0.55
L1	0.25	0.55	0.40
M	0.085	0.150	0.110
a	0°	8°	--
All Dimensions in mm			

Suggested Pad Layout

Please see <http://www.diodes.com/package-outlines.html> for the latest version.

SOT23



Dimensions	Value (in mm)
C	2.0
X	0.8
X1	1.35
Y	0.9
Y1	2.9

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